MS Brain MRI Viewer

A Cloud-Native Platform for Multiple Sclerosis Lesion Analysis, Longitudinal Tracking, and AI-Assisted Reporting

TECHNICAL DOCUMENTATION

Version 2.1 | September 2026

Classification: Medical Imaging Software

Research & Clinical Decision Support

Full-Stack Web Application | React + FastAPI | ~82,400 Lines of Code

Firebase Hosting | Google Cloud Run | NiiVue | Claude API | ONNX Runtime

Revision 2.1 (2026-09-28, CAPA-006, HAZ-005, RC-030): Section 5 corrected to the code. Version 2.0 (dated March 2026, committed 2026-04-02) described a Tier 2 (parcellation) / Tier 1 (MSMask atlas) hierarchy and a per-lesion "confidence"; it did not match the code when committed, which since 2026-03-14 has tried four paths in order (LST-AI zones, parcellation + EDT, MSMask atlas, geometric heuristics), and the confidence was uncalibrated and has been removed (RC-010 amended). Section 5 also states the CAPA-006 code changes of 2026-09-28: the MSMask atlas (adapted from LST-AI, not validated) is used only for images already registered to MNI152 that pass a five-condition grid check (no registration is performed; a stored zone map is never reused); the geometric heuristics are refused unless their image preconditions hold; when no path can be applied validly, no regions are assigned. Region-assignment accuracy has not been measured for any path. Section 6.1 corrected from 26-connectivity to 18-connectivity (RC-030).

1. ABSTRACT

This document presents the complete technical specification of the **MS Brain MRI Viewer**, a cloud-native web application designed for the visualization, segmentation, and quantitative analysis of brain magnetic resonance imaging (MRI) data in the context of Multiple Sclerosis (MS). The platform integrates real-time 2D/3D neuroimaging visualization with automated lesion detection, region-specific classification conforming to the MAGNIMS 2024 consensus guidelines, longitudinal disease progression tracking via IoU-based lesion matching, AI-powered clinical report generation through the Claude API, and edge-based neural network inference using ONNX Runtime Web.

The system architecture follows a decoupled client-server paradigm with a **React/TypeScript** frontend deployed on Firebase Hosting and a **FastAPI/Python** backend deployed on Google Cloud Run. State management employs Zustand with a single-source-of-truth pattern. The codebase comprises approximately **82,400 lines** of production code across **210+ files**, implementing features from low-level binary protocol optimization (17–42x throughput improvement) to high-level clinical decision support through natural language report generation.

2. SYSTEM ARCHITECTURE

2.1 High-Level Overview

The application follows a three-tier architecture: a single-page application (SPA) frontend served via Firebase Hosting CDN, a stateless API backend running on Google Cloud Run with auto-scaling, and a persistence layer combining Firestore (document metadata), Google Cloud Storage (binary NIfTI/DICOM blobs), Redis (response caching), and PostgreSQL (relational data).

2.2 Frontend Technology Stack

Layer	Technology	Version	Purpose
Framework	React	18.3.1	Component-based UI
Language	TypeScript	5.6.2	Type-safe development
Build	Vite	5.4.8	Fast HMR and bundling
State	Zustand	4.5.5	Lightweight reactive state
Data Fetching	TanStack React Query	5.56.2	Server state with caching
Medical Imaging	NiiVue	0.67.0	WebGL NIfTI volume rendering
3D Graphics	Three.js	0.169.0	WebGL scene management
Styling	Tailwind CSS	3.4.13	Utility-first CSS
i18n	i18next	25.6.3	Multi-language support
Edge ML	ONNX Runtime Web	1.21.0	Browser neural network inference

Table 1. Frontend technology stack with versions and primary responsibilities.

2.3 Backend Technology Stack

Layer	Technology	Version	Purpose
Framework	FastAPI	0.115.0	Async REST API
DICOM	pydicom	2.4.4	DICOM file parsing
NIfTI	nibabel	5.3.0	Neuroimaging file I/O
Image Processing	SimpleITK	2.3.1	Registration and filtering
Computer Vision	OpenCV	4.10.0	Image transformations
Scientific	NumPy/SciPy	1.26.4/1.13.1	Array ops, signal processing
AI Integration	Anthropic SDK	0.44.0	Claude API for reports
MCP	FastMCP	2.3.0	Model Context Protocol servers
Database	Firestore + PostgreSQL	—	Document and relational persistence
Storage	Google Cloud Storage	3.1.1+	Binary blob storage

Table 2. Backend technology stack.

2.4 State Management Architecture

The frontend employs five Zustand stores following the single-source-of-truth principle:

Store	LOC	Responsibility
useSegmentationStore	701	Segmentation masks, labels, paint tools, overlays, zone maps, longitudinal state
useViewerStore	171	Current series, slice index, zoom/pan, render mode, 3D settings
useMultiViewerStore	159	Multi-panel layout, synchronized slice navigation
usePatientStore	178	Patient context and current selection
useAIStore	94	AI task status and results

Table 3. Zustand state stores and their responsibilities.

3. MEDICAL IMAGE DATA PIPELINE

3.1 NIfTI Coordinate System

The system maintains two coordinate conventions with explicit bidirectional transformation:

```
Internal Convention (Backend Memory):
  Array shape: (D, H, W) = (Depth, Height, Width)
  Indexing:    mask[slice_z, row_y, col_x]

NIfTI File Convention:
  Array shape: (W, H, D) = (Width, Height, Depth)
  Orientation: RAS+ (Right, Anterior, Superior)
```

The affine matrix **A** transforms voxel coordinates **v** = (i, j, k, 1)^T to world coordinates **w** = (x, y, z, 1)^T:

$$\mathbf{w} = \mathbf{A} \cdot \mathbf{v} \quad \text{where } \mathbf{A} \in \blacksquare\blacksquare\blacksquare\blacksquare \text{ encodes rotation, scaling, and translation} \qquad \text{Eq. 1}$$

Conversion between conventions:

```
NIfTI → Internal:  masks_3d = np.transpose(nifti_data, (2, 1, 0))
Internal → NIfTI:  nifti_data = np.transpose(masks_3d, (2, 1, 0))
```

3.2 Binary Protocol Performance

A custom binary protocol replaces Base64 JSON encoding for slice data transmission, achieving significant throughput improvements:

Method	Encoding Time	Decoding Time	Bandwidth
Base64 JSON	12.4 ms	8.7 ms	1.33x original
Binary Protocol	0.3 ms	0.5 ms	1.00x original
Speedup	41.3x	17.4x	25% saving

Table 4. Binary protocol performance comparison vs Base64 encoding.

4. SEGMENTATION ENGINE

4.1 ITK-SNAP Local-First Architecture

The segmentation system follows an ITK-SNAP-inspired architecture where painting operations occur entirely in browser memory, eliminating network latency during interactive editing:

```
1. LOAD:  Download 3D binary mask once from server
2. PAINT: Modify local Uint8Array directly (instant feedback)
3. UNDO:  Slice-level snapshots stored in memory
4. SAVE:  Upload complete mask to server on demand
```

4.2 Paint Tools

Tool	Algorithm	Complexity
Brush (circle/square)	Bresenham circle fill with configurable radius	O(r²) per stroke point
Eraser	Identical to brush with label = 0	O(r²) per stroke point
Flood Fill	BFS from seed point with 4-connectivity	O(n), n = filled area
Threshold	Seed-based region growing within [min, max]	O(n)

Table 5. Paint tool algorithms and computational complexity.

4.3 MAGNIMS Regional Label Preset

ID	Name	Color	MAGNIMS Region
1	Periventricular	#FF4444 (Red)	Abutting lateral ventricles
2	Juxtacortical	#44BB44 (Green)	Touching cortical gray matter
3	Infratentorial	#4488FF (Blue)	Brainstem or cerebellum
4	Deep White Matter	#FFAA00 (Yellow)	Centrum semiovale, corona radiata
5	Active (Gd+)	#FF00FF (Magenta)	Gadolinium-enhancing
6	Black Hole (T1)	#8844CC (Purple)	Chronic destructive

Table 6. MAGNIMS regional label preset with color encoding.

5. MAGNIMS REGION CLASSIFICATION

Criterion (MAGNIMS / McDonald 2024): PV = abutting the lateral ventricles, JC = abutting the cortex, IT = in or touching the brainstem/cerebellum, otherwise DWM. On the parcellation, MSMask and geometric paths the priority is IT > PV > JC > DWM; the LST-AI path copies LST-AI's zones voxel by voxel. Only PV, JC and IT count towards brain DIS (spinal cord and optic nerve are not evaluated from the images; clinician-entered evidence for them is accepted by the DIS assessment). In auto mode the paths are tried in order and the first that applies is used; the path used is reported. If no path can be applied validly, the request is refused (HTTP 422) with the reasons and no regions are assigned. Region-assignment accuracy has not been measured for any path (HAZ-005 residual undetermined).

- **Path 1 — LST-AI zones:** only if another segmentation of the same image has a validation_source containing "lst-ai" (LST-AI integration disabled by default, LSTAI_ENABLED=false; not enabled by the deployment pipeline). Copies, voxel by voxel, the zone LST-AI assigned: no contact rule and no priority, so one lesion can receive more than one region; lesion voxels outside LST-AI's zones become DWM without a warning.
- **Path 2 — Parcellation + EDT** (Section 5.1): with an explicit parcellation_id, or when another segmentation of the same image looks like a FreeSurfer parcellation (at least 3 of the labels {2,3,4,7,8,10,16,41,42,43} and at least one of {16,41,42,43}, and not a MAGNIMS zone map; an explicit parcellation_id that fails the test is refused, HTTP 422). SynthSeg integration disabled by default (SYNTHSEG_ENABLED=false); when enabled it runs as a separate service at SYNTHSEG_ENDPOINT, not on Google Vertex AI.
- **Path 3 — MSMask atlas** (Section 5.3; adapted from LST-AI, not validated): only for images already registered to MNI152; no registration is performed. Native clinical scans normally fail its grid check and are then classified by path 4, or refused.
- **Path 4 — Geometric heuristics** (Section 5.4; coordinate rules, not anatomical landmarks; accuracy not measured): refused unless its image preconditions hold.

5.1 Path 2: Parcellation-Based EDT Classification

This path uses the Euclidean Distance Transform (EDT) computed from SynthSeg/FreeSurfer parcellation volumes of the same image. For each anatomical reference structure, a binary mask is extracted from the parcellation using FreeSurfer label groups:

Structure	FreeSurfer Labels	Role
Lateral Ventricles	{4, 43}	PV reference
Cerebral Cortex	{3, 42}	JC reference
Infratentorial	{7, 8, 16, 46, 47}	IT reference
Cerebral White Matter	{2, 41}	DWM default

Table 7. FreeSurfer label groups for MAGNIMS region classification.

The EDT is computed for each reference mask, yielding distance fields in millimeters:

$$D_{vent}(x) = \min_{y \in M_{vent}} ||x - y||_2 \cdot \mathrm{diag}(\Delta) \quad \text{Eq. 2}$$

where $\Delta = (\Delta x, \Delta y, \Delta z)$ is the voxel spacing vector.

For each connected component C_k, the minimum distance to each reference structure is computed, and classification follows a **priority cascade** (IT > PV > JC > DWM):

```
IF d_IT <= 1.5 mm:     region = Infratentorial
ELIF d_PV <= 1.5 mm:  region = Periventricular
ELIF d_JC <= 1.5 mm:  region = Juxtacortical
ELSE:                 region = Deep White Matter
```

Distances are between voxel centres. Limits of the 1.5 mm rule: with slices thicker than 1.5 mm, contact through the slice direction is not detected; at 1 mm voxels, diagonal (corner) neighbours at 1.73 mm are missed.

5.2 Evidence Reported (No Per-Lesion Confidence)

No per-lesion confidence is emitted or displayed (confidence is always null; HAZ-005, REQ-SAFE-010, RC-010 (amended)). Region assignment is a deterministic MAGNIMS rule. The parcellation path reports distances_mm (d_IT, d_PV, d_JC; null when a landmark is absent) as evidence.

5.3 Path 3: MSMask Atlas Zone Map (MNI-space images only)

The atlas-based method is adapted from LST-AI's MSMask atlas (Wiltgen et al. 2024); the adaptation used here has not been validated. The atlas is in MNI152 space and MSTool-AI performs no registration (the zone map is resampled onto the image grid only), so it is used only for images already registered to MNI152 whose grid passes all five conditions: (1) isotropic voxels (max/min spacing at most 1.05); (2) axes aligned with and not permuted relative to the world axes (within about 2.5°); (3) field of view within 25% of 181 × 217 × 181 mm on each axis; (4) grid centre within 10 mm of the MNI template centre (0, -18, 18) mm; (5) a spatial transform in the header (sform or qform code not 0). Otherwise it is refused (HAZ-005, RC-032): HTTP 422 for an explicit MSMask request; in auto the geometric heuristics are tried and the UI shows "Atlas regions not used: this image is not in MNI space, so the geometric fallback (coordinate rules, not anatomical landmarks) was applied. Verify every region before using it for DIS." The check cannot certify true normalization (an image resampled onto an MNI grid without registration passes). The check is on the grid, not on registration: an image resampled onto an MNI grid without registration, or a well-behaved native 3D isotropic scan with an MNI-like field of view and centre, can pass it; SPM12 'Normalise: Write' outputs with the default bounding box are refused (their grid centre is about 10.5 mm from the template centre). The zone map is generated fresh for every classification; a stored zone map is never reused. Zones: the atlas ventricle, cortex and infratentorial structures dilated by one voxel (3×3×3 cube on the 1 mm atlas) and intersected with atlas white matter, plus the infratentorial structures themselves; zone precedence IT > PV > JC:

```
zone_map[WM] = 4                                    # DWM default
zone_map[IT or IT_dilated AND WM] = 3              # Infratentorial
zone_map[Cortex_dilated AND WM AND zone != 3] = 2  # Juxtacortical
zone_map[Vent_dilated AND WM AND zone != 3] = 1    # Periventricular
```

A lesion is assigned the highest-priority zone that any of its voxels lies in (IT > PV > JC > DWM); a lesion with no voxel in any zone is Deep White Matter by default. Per-lesion evidence (no confidence): region_overlap_fraction (fraction of all lesion voxels inside the assigned zone), zone_coverage_fraction (fraction inside any white-matter zone) and atlas_coverage (false when the lesion lies in no white-matter zone: Deep White Matter is then assigned by default, not by the MAGNIMS rule, and flagged in the UI as "No WM zone — DWM by default").

5.4 Path 4: Geometric Heuristics (accuracy not measured)

Coordinate rules, not anatomical landmarks. The path is refused (HTTP 422 for an explicit request; in auto, the request is refused with the reasons when no other path applies) unless the source image can be read, its orientation can be determined, its slice axis runs from inferior to superior (axial acquisition; sagittal, coronal or superior-to-inferior slice order is refused), it is tilted no more than 30° from that axis, it is 3-D, it is on the same grid as the lesion mask and it is not blank. Brain outline = Otsu threshold (x0.3) of the image intensities, holes filled (the head outline if the image is not skull-stripped). IT = lesion centroid in the lowest 25% of the outline's extent along the slice axis; PV = mean distance of the lesion's voxels from the brain centre (midpoint of the outline's slice extent and the in-plane array centre) below 35% of the largest such distance in the image array; JC = lesion within min(8 mm, 15% of the maximum depth of the outline) of the outline surface; otherwise DWM; priority IT > PV > JC > DWM. Whenever regions come from this path the UI shows "Geometric heuristics: coordinate rules, not anatomical landmarks; their accuracy has not been measured. Verify every region before using it for DIS." (or the atlas warning above when the atlas was refused).

6. LESION ANALYSIS AND DIS ASSESSMENT

6.1 Connected Component Extraction

Lesion components are extracted using scipy.ndimage.label() with 18-connectivity (faces + edges; RC-030). For each component C_k:

$$\mathrm{vol}(C_k) = |C_k| \cdot \Delta x \cdot \Delta y \cdot \Delta z \quad [\mathrm{mm}^3] \tag{Eq. 5}$$

$$\mathbf{x}_k = \frac{1}{|C_k|} \sum_{x \in C_k} x \quad [\text{centroid}] \tag{Eq. 6}$$

Components with volume < 3.0 mm³ are discarded as noise. Size categories: **small** (< 100 mm³), **medium** (100–1000 mm³), **large** (> 1000 mm³).

6.2 McDonald 2024 DIS Criteria

The Dissemination in Space (DIS) assessment follows the revised McDonald criteria (Montalban et al., 2025). The brain-only assessment evaluates three of five regions:

$$\mathrm{DIS}_{\mathrm{brain}} = \mathbb{1}\left[\sum_{r \in \{PV,\ JC,\ IT\}} \mathbb{1}\left[\exists\, C_k : \mathrm{region}(C_k) = r \wedge \mathrm{vol}(C_k) \geq 3.0\right] \geq 2\right] \tag{Eq. 7}$$

DIS is met when at least 2 of the 3 evaluable brain regions contain qualifying lesions (volume ≥ 3.0 mm³).

7. LONGITUDINAL TRACKING

7.1 IoU-Based Lesion Matching

Given two timepoint masks M_{t1} and M_{t2}, connected components are extracted independently and matched using the Intersection over Union metric:

$$\mathbf{IoU(A_i, B_j) = |A_i \cap B_j| / |A_i \cup B_j|} \quad \text{Eq. 8}$$

A greedy bipartite matching algorithm pairs components with IoU ≥ 0.3:

```
FOR i = 1 TO p:  (TP1 components)
    best_j = argmax_{j not used} IoU(A_i, B_j)
    IF IoU(A_i, B_best_j) >= 0.3:
        matches.append((A_i, B_best_j, IoU))
        mark B_best_j as used

unmatched_A = resolved lesions
unmatched_B = new lesions
```

7.2 Status Classification

For each matched pair (A_i, B_j):

$$\delta = (vol(B_j) - vol(A_i)) / vol(A_i) \times 100\% \quad \text{Eq. 9}$$

Condition	Status	Clinical Meaning
δ > 20%	Enlarged	Lesion growth indicating activity
δ < -20%	Shrunk	Lesion regression
|δ| ≤ 20%	Stable	No significant change
A_i unmatched	Resolved	Lesion disappeared
B_j unmatched	New	New lesion appeared

Table 8. Longitudinal lesion status classification criteria.

7.3 Visual Overlay

The longitudinal comparison renders a tri-color overlay on the 2D canvas and 3D NiiVue viewer:

Condition	Color	RGBA	Meaning
Voxel in TP1 only	Blue	(65, 135, 245, 140)	Resolved / decreased
Voxel in TP2 only	Red	(245, 70, 70, 140)	New / increased
Voxel in both TPs	Green	(50, 205, 50, 140)	Persistent

Table 9. Longitudinal overlay color encoding.

8. BRAIN VOLUMETRY

8.1 Volume Calculation

$$V_l = |\{(i,j,k) : P(i,j,k) = l\}| \cdot \Delta x \cdot \Delta y \cdot \Delta z \quad [mm^3]$$ Eq. 10

8.2 Normative Percentile (Z-Score)

Given measured volume V_l and normative statistics (μ_l, σ_l) for the patient's age group:

$$z = (V_l - \mu_l) / \sigma_l$$ Eq. 11

$$percentile = 50 \cdot (1 + erf(z / \sqrt{2}))$$ Eq. 12

where erf is the Gauss error function. Age groups: 20–40, 40–60, 60–80, 80+.

8.3 Abnormality Detection

Structure Type	FreeSurfer Labels	Abnormality Criterion
Ventricular	{4, 43, 5, 44, 14, 15}	Percentile > 90 (enlargement)
Atrophy-sensitive	{17, 53, 10, 49, 11, 50, 12, 51}	Percentile < 10 (atrophy)

Table 10. Abnormality detection criteria by structure type.

9. SEGMENTATION COMPARISON METRICS

9.1 Dice Similarity Coefficient

$$DSC(A, B) = 2|A \cap B| / (|A| + |B|)$$ Eq. 13

Properties: DSC ∈ [0, 1]; DSC = 1 indicates perfect overlap. By convention, DSC(∅, ∅) = 1.

9.2 Hausdorff Distance (95th Percentile)

The directed surface distance from set A to set B:

$$d(a, B) = \min_{b \in \partial B} ||a - b||_2 \cdot diag(\Delta)$$ Eq. 14

$$HD_{95}(A, B) = \max(P_{95}\{d(a,B) : a \in \partial A\}, \; P_{95}\{d(b,A) : b \in \partial B\})$$ Eq. 15

Implementation uses EDT for efficient surface distance computation in O(n) time where n is the volume size (Maurer et al., 2003).

10. 3D VISUALIZATION PIPELINE

10.1 NiiVue Integration

The 3D viewer uses NiiVue (v0.67.0), a WebGL2-based neuroimaging viewer operating in two modes:

- **Volume Mode:** Single canvas with ray-casting rendering
- **Multiplanar Mode:** 2×2 synchronized grid (3D + axial + coronal + sagittal)

10.2 Overlay Layer Stack

Index	Layer	Colormap	Default Opacity
0	Main MRI	User-selected	1.0
1	Segmentation	red	0.5
2	Zone Map	Custom magnims_zones	0.0–1.0 (reactive)
3	Longitudinal TP1	blue	0.5
4	Longitudinal TP2	hot	0.5

Table 11. NiiVue overlay layer stack with rendering order.

11. 2D RENDERING ENGINE

11.1 Canvas Layer Architecture

The SegmentationCanvasLocal component (970 lines) implements a multi-layer rendering pipeline with 8 composited layers:

Layer	Content	Opacity Control
0	Base MRI Image	Fixed (1.0)
1	Zone Map Background (MAGNIMS zones)	zoneMapOpacity slider
2	Longitudinal Overlay (TP1/TP2/overlap)	Fixed (55%)
3	Segmentation Mask (label colors)	lesionOpacity + per-label visibility
4	Heatmap (anomaly probability)	Value-proportional
5	AI Click Points	Fixed markers
6	Selected Lesion (bbox + centroid)	Fixed highlight
7	Cursor Preview (brush outline)	Mouse-following

Table 12. Canvas rendering layer stack (bottom to top).

11.2 Heatmap Colormap (Hot)

For a normalized value $t \in [0, 1]$:

```
t < 0.33:      RGB = (t/0.33 * 255,  0,  0)           Black → Red
0.33 <= t < 0.66: RGB = (255,  (t-0.33)/0.33 * 255,  0)  Red → Yellow
t >= 0.66:        RGB = (255,  255,  (t-0.66)/0.34 * 255)  Yellow → White
alpha = t * 200
```

12. AI-ASSISTED REPORT GENERATION

12.1 HIPAA-Compliant Pipeline

The report generation system integrates with the Claude API (Anthropic SDK v0.44.0) through a de-identification pipeline that strips all Protected Health Information (PHI) before transmission:

```
Clinical Findings → De-identification → Prompt Assembly → Claude API → Report
```

Template	Max Tokens	Clinical Focus
general	4,096	Standard MS brain MRI report
ms_activity	4,096	Disease activity assessment
ms_lesion_burden	6,144	Quantitative lesion inventory
ms_comprehensive	8,192	Publication-quality comprehensive report

Table 13. Report generation templates and token budgets.

12.2 De-Identification Protocol

Transmitted: Patient age (integer), sex (M/F), clinical indication, technical parameters, numerical findings (lesion counts, volumes, regions), DIS assessment, longitudinal statistics.

Never transmitted: Patient name, date of birth, MRN, study date, institution name, referring/performing physician names.

13. EDGE AI INFERENCE

The Edge AI module performs neural network inference directly in the browser using ONNX Runtime Web with WebGPU acceleration (WASM fallback):

```
Image Slice (Uint8Array) → Web Worker → ONNX Runtime (WebGPU/WASM) → Classification
```

Preprocessing: bilinear resize to 224×224, min-max normalization, Float32Array tensor [1, 1, 224, 224]. Output: softmax for multi-class or sigmoid for binary classification (Normal/Abnormal).

14. BINARY PROTOCOL SPECIFICATION

14.1 Message Header (24 bytes, Little-Endian)

Offset	Field	Type	Size	Description
0–3	magic	uint32	4	0x4D4449 ("MDI")
4–5	version	uint16	2	Protocol version (1)
6	message_type	uint8	1	Message type enum
7	compression	uint8	1	Compression algorithm
8–11	payload_length	uint32	4	Payload size in bytes
12–15	sequence_num	uint32	4	Monotonic packet counter
16–19	crc32	uint32	4	CRC32 of payload
20–23	reserved	uint32	4	Reserved for future use

Table 14. Binary protocol message header format.

14.2 CRC32 Implementation

The CRC32 checksum uses the standard IEEE 802.3 polynomial 0xEDB88320 (reflected form):

$$\text{CRC}(M) = \oplus_{i=0}^{|M|-1} \; T[(\text{CRC}_i \oplus M_i) \;\&\; \text{0xFF}] \oplus (\text{CRC}_i \gg 8) \qquad \text{Eq. 16}$$

15. MODEL CONTEXT PROTOCOL

Five MCP servers provide Claude with 22 specialized tools for medical imaging analysis:

Server	Tools	Purpose
imaging_server	4	Image metadata, slice retrieval, matplotlib rendering
segmentation_server	5	AI segmentation, volumetry, anomaly detection
report_server	3 + 3 resources	Report generation, templates, differential diagnosis
ms_clinical_server	6	MS-specific clinical data and assessment
pacs_server	4	PACS / study data access

Table 15. MCP server architecture with tool counts.

16. SECURITY ARCHITECTURE

Layer	Mechanism	Implementation
Authentication	Firebase Auth + JWT	Token refresh via Axios interceptors
Encryption	AES-256-GCM	PBKDF2 key derivation (100k iterations)
Audit	HIPAA-compliant logging	User, action, resource, timestamp per access
Transport	TLS 1.3	Enforced for all communications
Data at Rest	AES-256	GCS server-side encryption
Rate Limiting	Token bucket	100 req capacity, 10 req/s refill
Input Validation	Pydantic + custom	Schema validation and sanitization

Table 16. Multi-layer security architecture.

17. PERFORMANCE OPTIMIZATION

Technique	Component	Impact
Binary Protocol	Data transfer	17–42x throughput vs Base64
Two-Tier Cache	Memory + IndexedDB	<1ms (L1) / 2–10ms (L2) access
Canvas Pooling	Rendering	Eliminates GC during slice navigation
Virtual Scrolling	Patient/study lists	O(visible) render, not O(total)
Web Workers	AI inference + binary decode	Non-blocking main thread
Transferable Buffers	Worker communication	Zero-copy ArrayBuffer transfer
LRU Eviction	Cache management	Bounded memory with optimal hit rate

Table 17. Performance optimization techniques and measured impact.

18. INTERNATIONALIZATION

The application supports three languages with complete coverage:

Code	Language	Keys	Coverage
en	English	1,160	100%
es	Spanish	1,160	100%
de	German	1,155	100%

Table 18. Internationalization coverage.

19. DEPLOYMENT ARCHITECTURE

Service	Provider	Configuration
Frontend Hosting	Firebase Hosting	Global CDN, SPA routing, HTTPS
Backend Compute	Cloud Run	Auto-scaling 0–N, 2 GiB RAM, 2 vCPU
Database	Firestore	Document metadata, patient records
Blob Storage	Cloud Storage	NIfTI/DICOM files, segmentation masks
Cache	Redis (Memorystore)	Response caching
AI/ML	Vertex AI + Anthropic	Model endpoints + report generation

Table 19. Cloud infrastructure deployment map.

20. KEY CONSTANTS REFERENCE

Constant	Value	Unit	Context
MIN_LESION_VOLUME	3.0	mm^3	Noise filter threshold
PV_DISTANCE_THRESHOLD	1.5	mm	Periventricular EDT
JC_DISTANCE_THRESHOLD	1.5	mm	Juxtacortical EDT
IT_DISTANCE_THRESHOLD	1.5	mm	Infratentorial EDT
IoU_THRESHOLD	0.3	—	Longitudinal match
CHANGE_THRESHOLD	0.20	—	Volume change (20%)
DIS_BRAIN_REGIONS	3	count	Evaluable brain regions
DIS_TOTAL_REGIONS	5	count	Full McDonald 2024
LESION_OPACITY_DEFAULT	60	%	Overlay transparency
ZONE_MAP_OPACITY_DEFAULT	30	%	Background zone opacity

Table 20. Key constants and clinical thresholds used throughout the system.

. REFERENCES

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A. APPENDIX: CODEBASE STATISTICS

Metric	Value
Total files	210+
Total lines of code	~82,400
Frontend files / LOC	119 / ~40,200
Backend files / LOC	87 / ~38,800
Test files / LOC	11 / ~5,300
TypeScript type definitions	1,330 lines
i18n translation keys	1,160 per language
API endpoints	~60
Zustand stores	5
React hooks	22
Backend services	24
MCP tools	22

Table A.1. Complete codebase statistics.